

Four-Corner Spectral Tuning for Quadratic ADMM with Application to Diffeomorphic Image Registration

Katherine Xie Jie Wu Xiaohui Xie*

Department of Computer Science, University of California, Irvine, Irvine, California, USA

Abstract

Selecting the penalty and over-relaxation parameters of quadratic ADMM can require numerical optimization of the spectral radius of a large iteration operator. We identify a separable class for which this optimization reduces to four scalar modes. When the joint modal spectrum is a Cartesian product of data and regularizer curvatures with attained endpoint combinations, the oADMM modal factor is coordinatewise monotone, so its worst magnitude is attained at four endpoint combinations. The conditional relaxation has a closed form, and the remaining penalty problem is solved by a finite algebraic candidate set rather than a grid search, gradient descent, or repeated spectral-radius calculation. Spatially varying diffeomorphic registration is noncommuting and does not satisfy the exact theorem globally. Local coefficient freezing recovers the separable structure, while the rank-one optical-flow Hessian yields an inexpensive four-corner parameter predictor. On 134 FIRE retinal pairs at 256×256 , one-shot tuning reduced median pairwise runtime by 34.4% and inner iterations by 39.6% relative to a fixed pair selected on separate validation data, with essentially unchanged landmark error and no observed nonpositive Jacobian. On ten CIMA histology pairs, matched-protocol runtime decreased by 20.1%. Optional noncommuting rate bounds are rigorous but conservative and are not used by the deployed selector.

Keywords: alternating direction method of multipliers; parameter selection; spectral analysis; diffeomorphic image registration; local Fourier analysis

1 Introduction

Alternating direction methods are attractive for large quadratic imaging problems because each split subproblem can retain its natural computational structure. In variational image registration, for example, one update is pixelwise and the other is diagonal in a Fourier basis [14, 12]. This advantage does not remove the sensitivity of ADMM to its penalty ρ and, for over-relaxed ADMM (oADMM), its relaxation α . Poor fixed values can dominate the cost of an otherwise efficient solver.

Optimal parameter formulas are known for several quadratic classes [5, 13, 6]. Song et al. treat a more general linear-quadratic problem by mapping each trial penalty to an iteration operator, evaluating its spectral radius, and numerically optimizing over the penalty; relaxation is then selected conditionally [12]. The cost is *many trial values of ρ times spectral-radius evaluation of a large operator*. The penalty does not change during a fixed inner solve: the expense is incurred while finding the fixed pair for that quadratic subproblem.

The central observation here applies to a separable quadratic split. Its oADMM contraction on a mode depends on only two nonnegative curvatures, h and g . The resulting scalar function is

*Corresponding author: xhx@uci.edu.

monotone in each coordinate on a spectral rectangle. Consequently, its worst magnitude is attained at the four corners. This reduces a large spectral optimization to four rational branches. The relaxation can be eliminated analytically, and a finite set of algebraic candidates contains a global minimizer of the resulting four-corner objective on a prescribed admissible penalty interval.

Real image registration does not possess this global separability: the pixelwise image Hessian varies spatially, while the regularizer couples neighboring pixels. We recover the exact modal model locally by freezing the image coefficient. The rank-one optical-flow Hessian then collapses all local curvature endpoints to two global values. After an $O(n)$ scan of the image gradients, parameter optimization has only four scalar branches. The conceptual sequence is therefore

$$\begin{aligned} \text{quadratic ADMM} &\longrightarrow \text{modal factor} \longrightarrow \text{four corners} \\ &\longrightarrow \text{finite parameter selection} \longrightarrow \text{noncommuting registration.} \end{aligned}$$

The contributions are:

1. a four-corner spectral theorem for quadratic oADMM with a Cartesian-product joint modal spectrum;
2. a finite algebraic method for jointly selecting ρ and α for that model, without repeated iteration-matrix eigensolves;
3. a local-freezing extension to spatially varying problems and a rank-one specialization that gives a one-shot registration predictor; and
4. matched-protocol FIRE and CIMA experiments showing reduced iterations and runtime with essentially unchanged registration accuracy and no observed nonpositive discrete Jacobian under the reported protocol.

2 Related Work

2.1 ADMM parameter selection

Scaled ADMM and practical residual criteria were systematized by Boyd et al. [2]. Ghadimi et al. derived optimal parameters for several quadratic classes [5]; Teixeira et al. studied parameter selection, scaling, and preconditioning in distributed quadratic programs [13]; and Giselsson and Boyd developed tight metric-dependent contraction bounds [6]. Adaptive alternatives include residual balancing and Barzilai–Borwein (BB) spectral updates [17, 18].

2.2 Spectral and over-relaxed ADMM

Song et al. optimize ADMM and oADMM parameters for general linear-quadratic problems [12]. Their pipeline is

$$\rho \longmapsto E_\rho \longmapsto \varrho(E_\rho) \longmapsto \text{numerical optimization over } \rho,$$

with an optimal or closed-form relaxation conditional on ρ . In contrast, the present separable model maps four spectral endpoints to four rational branches and then to a finite algebraic candidate set. This distinction concerns parameter-selection overhead and structure, not changing ρ during inner ADMM iterations. We do not claim superiority to full-resolution direct spectral optimization without a direct experiment at that resolution.

2.3 Fourier and local Fourier analysis

Fourier analysis has long characterized stationary iterations for elliptic problems [4], and local Fourier analysis (LFA) is a standard tool for multigrid design and parameter optimization [16]. Translation-invariant periodic problems admit exact Fourier decompositions; local and nonstandard analyses can remain informative for random or discontinuous coefficients [11, 9]. Here the stationary map is an ADMM error iteration, and local freezing converts a matrix-valued optical-flow coefficient into a separable patch model.

2.4 Variational image registration

DARTEL [1], diffeomorphic demons [15], and large-deformation optimal-control formulations [10] are representative optimization-based registration methods. Thorley et al. compose nonstationary velocity updates and solve regularized convex subproblems with accelerated ADMM on GPUs [14]. Song et al. use a related quadratic model in their spectral oADMM study [12]. Learning-based methods broaden the current landscape [7], but the present work concerns numerical parameter selection for the linearized variational subproblem.

3 Quadratic ADMM and Reduced Error Dynamics

3.1 Quadratic splitting

Consider

$$\min_{v,w} \underbrace{\frac{1}{2}v^\top H v + b^\top v}_{f(v)} + \underbrace{\frac{1}{2}w^\top G w}_{r(w)} \quad \text{subject to} \quad v - w = 0, \quad (1)$$

where $v, w, b \in \mathbb{R}^N$ and $H, G \in \mathbb{R}^{N \times N}$ are symmetric positive semidefinite matrices. The linear term b determines the fixed point but disappears from homogeneous error propagation.

3.2 Standard and over-relaxed ADMM

With scaled dual variable u and penalty $\rho > 0$, standard ADMM is

$$(H + \rho I)v^{k+1} = -b + \rho(w^k - u^k), \quad (2)$$

$$(G + \rho I)w^{k+1} = \rho(v^{k+1} + u^k), \quad (3)$$

$$u^{k+1} = u^k + v^{k+1} - w^{k+1}. \quad (4)$$

For over-relaxed ADMM, abbreviated *oADMM*, define

$$\hat{v}^{k+1} = \alpha v^{k+1} + (1 - \alpha)w^k, \quad 0 < \alpha \leq 2, \quad (5)$$

and replace v^{k+1} by $\hat{v}^{k+1}$ in (3) and the dual update.

3.3 Reduced Cayley representation

Define

$$C_H(\rho) = (H - \rho I)(H + \rho I)^{-1}, \quad C_G(\rho) = (G - \rho I)(G + \rho I)^{-1}.$$

Proposition 3.1 (Reduced Cayley error operator). *The nonzero spectra of the oADMM state map and*

$$E_{\rho,\alpha} = \left(1 - \frac{\alpha}{2}\right) \mathbf{I} + \frac{\alpha}{2} C_H(\rho) C_G(\rho). \quad (6)$$

coincide, including algebraic multiplicity; the remaining state-map eigenvalues are zero. Thus the asymptotic factor is $\varrho(E_{\rho,\alpha})$, where $\varrho(\cdot)$ denotes the spectral radius of a matrix or linear operator.

Appendix A gives the reduction. It is supporting machinery: the parameter reduction below comes from the modal dependence of (6).

Theorem 3.2 (Basic contraction). *If $H, G \succeq 0$ and $H + G \succ 0$, then for every $\rho > 0$, $\kappa_\rho := \|C_H(\rho)C_G(\rho)\|_2 < 1$. Consequently*

$$\varrho(E_{\rho,\alpha}) \leq 1 - \frac{\alpha}{2}(1 - \kappa_\rho) < 1,$$

for $0 < \alpha \leq 2$.

Proof. For $A \succeq 0$, the Cayley eigenvalue $(\lambda - \rho)/(\lambda + \rho)$ has magnitude at most one, with equality only for $\lambda = 0$. Equality in $\|C_H C_G x\| \leq \|C_G x\| \leq \|x\|$ would imply $x \in \ker G$ and $C_G x = -x \in \ker H$, contradicting $H + G \succ 0$. Compactness gives strict inequality; the displayed estimate follows from (6). $\square$

Nullspaces. If a nonzero vector lies in $\ker H \cap \ker G$, then $C_H C_G x = x$ and $E_{\rho,\alpha} x = x$ for every admissible parameter pair. Strict contraction therefore requires $H + G \succ 0$, or restriction to the identifiable subspace. In the four-corner registration model, $\nu = \gamma = 0$ creates the $(h, g) = (0, 0)$ mode; the selector then requires positive damping, a coercive boundary/regularization condition, or explicit removal of that common nullspace.

3.4 Scalar modal factor

When a vector is a common eigenvector of H and G , with eigenvalues h and g , (6) acts by

$$\begin{aligned} e(h, g; \rho, \alpha) &= 1 - \frac{\alpha}{2} + \frac{\alpha}{2} \frac{h - \rho}{h + \rho} \frac{g - \rho}{g + \rho} \\ &= 1 - \alpha + \alpha \theta(h, g; \rho), \end{aligned} \quad (7)$$

$$\theta(h, g; \rho) = \frac{\rho^2 + hg}{(\rho + h)(\rho + g)}. \quad (8)$$

For nonnegative h and g , $0 \leq \theta \leq 1$. The paired values (h, g) , not the marginal spectra separately, determine the true modal factor.

4 Four-Corner Spectral Tuning

4.1 Separable commuting modal structure

Definition 4.1 (Cartesian-product joint modal spectrum). A quadratic split is *separable* on modal sets $\mathcal{H}$ and $\mathcal{G}$ if H and G are simultaneously diagonalizable and their attained joint modal spectrum is

$$\{(h, g) : Hx = hx, Gx = gx \text{ for a common mode } x \neq 0\} = \mathcal{H} \times \mathcal{G}.$$

For example, $H_{\text{sep}} = I_m \otimes H_c$ and $G_{\text{sep}} = G_s \otimes I_d$ have modes $\phi_\xi \otimes z_j$ and pairs $(h_j, g(\xi))$, where $H_c z_j = h_j z_j$, $G_s \phi_\xi = g(\xi) \phi_\xi$, $H_c \in \mathbb{R}^{d \times d}$ acts on components, $G_s \in \mathbb{R}^{m \times m}$ acts on spatial degrees of freedom, and I_m, I_d are the corresponding identity matrices.

The Cartesian-product condition is stronger than commutativity. It asserts that data and regularizer modes combine independently.

4.2 Four-corner theorem

For fixed $\rho > 0$,

$$\frac{\partial \theta}{\partial h} = \frac{\rho(g - \rho)}{(g + \rho)(h + \rho)^2}, \quad \frac{\partial \theta}{\partial g} = \frac{\rho(h - \rho)}{(h + \rho)(g + \rho)^2}. \quad (9)$$

The sign of the first derivative is independent of h , and the sign of the second is independent of g . Thus θ is coordinatewise monotone on any rectangle.

Theorem 4.2 (Four-Corner Spectral Theorem). *Suppose the joint modal spectrum is the Cartesian product $\mathcal{H} \times \mathcal{G}$, where $\mathcal{H} \subset [h_-, h_+]$ and $\mathcal{G} \subset [g_-, g_+]$, with $0 \leq h_- \leq h_+$ and $0 \leq g_- \leq g_+$. If all four endpoint combinations are attained, then*

$$\varrho(E_{\rho, \alpha}) = U_{4C}(\rho, \alpha) := \max_{\substack{h \in \{h_-, h_+\} \\ g \in \{g_-, g_+\}}} |e(h, g; \rho, \alpha)| \quad (10)$$

for every $\rho > 0$ and $0 < \alpha \leq 2$. If only the interval enclosure is known, the right-hand side remains a valid rectangular upper envelope, but equality with the true spectral radius need not hold.

Proof. Equation (9) makes the extrema of θ on the rectangle occur at corners. The map $t \mapsto |1 - \alpha + \alpha t|$ is convex, so its maximum over the interval between those extrema also occurs at an endpoint and hence at a corner. Cartesian-product structure and attained endpoint combinations make those corners actual modes, proving equality. $\square$

4.3 Optimal relaxation for fixed penalty

Let the four corner values satisfy

$$a(\rho) = \min_c \theta_c(\rho), \quad b(\rho) = \max_c \theta_c(\rho).$$

The four-corner objective is

$$\max\{|1 - \alpha + \alpha a|, |1 - \alpha + \alpha b|\}.$$

Equioscillation of the two endpoint factors gives the unconstrained optimizer; if $(a, b) \neq (1, 1)$, accounting for $0 < \alpha \leq 2$ yields

$$\alpha^*(\rho) = \min \left\{ 2, \frac{2}{2 - a(\rho) - b(\rho)} \right\}. \quad (11)$$

After eliminating α , the reduced objective is

$$\begin{aligned} \widehat{U}(\rho) &= \frac{b - a}{2 - a - b} && \text{if } a + b \leq 1, \\ \widehat{U}(\rho) &= 2b - 1 && \text{if } a + b \geq 1. \end{aligned} \quad (12)$$

In the degenerate case $a = b = 1$, the objective equals one for every admissible α ; any relaxation may be selected, but the mode is not contractive.

4.4 Finite algebraic penalty selection

First deduplicate identical rational corner branches $\theta_c(\rho)$. On a compact admissible interval $\mathcal{I} = [\rho_-, \rho_+] \subset (0, \infty)$, changes in the active minimum or maximum occur only at pairwise branch intersections.

Theorem 4.3 (Finite penalty candidate set). *A global minimizer of (12) over $\mathcal{I}$ is contained in the finite set formed by:*

1. ρ_- and ρ_+ ;
2. positive pairwise branch-intersection roots in $\mathcal{I}$;
3. roots of $a(\rho) + b(\rho) = 1$ for every active branch pair; and
4. stationary roots of each active rational expression in (12).

Identically zero intersection, transition, or stationarity polynomials are discarded; a constant objective piece is represented by its interval endpoints. Thus the candidate set remains finite when endpoints coincide. After positive denominators are cleared, the respective polynomial degrees are at most three, four, and six. Evaluating (11) and (10) on this set—using any admissible α in the degenerate $a = b = 1$ case—returns a global minimizer of the four-corner objective over $\mathcal{I} \times (0, 2]$.

The proof is in Appendix C. The result replaces a continuum of trial penalties by a constant number of polynomial roots:

no grid search + no gradient descent in ρ + no repeated spectral-radius calculation.

4.5 Arbitrary commuting operators versus Cartesian products

Three cases must be distinguished.

Case A: For a separable commuting split with attained endpoint combinations, (10) is the true spectral radius, and finite tuning is globally optimal for that model over $\mathcal{I} \times (0, 2]$.

Case B: If $HG = GH$ but the paired spectrum is only $\{(h_i, g_i)\}_i$, the exact factor is $\max_i |e(h_i, g_i; \rho, \alpha)|$. The four corners of the marginal spectral rectangle can be artificial and provide an envelope rather than an equality.

Case C: If $HG \neq GH$, no global scalar modal decomposition follows. Four-corner tuning is a surrogate or predictor unless a separate argument supplies a bound.

5 Spatially Varying and Noncommuting Problems

5.1 Why global modal analysis fails

Let $H = \text{diag}(H_1, \dots, H_n)$ vary by location and let $G = G_0 \otimes I_d$ couple neighboring sites. In block form,

$$[H, G]_{ij} = (G_0)_{ij}(H_i - H_j),$$

so $HG \neq GH$ whenever coupled sites have different data blocks. The exact factor is then

$$R_{\text{true}}(\rho, \alpha) = \varrho(E_{\rho, \alpha}),$$

not a maximum of the scalar factors in (7).

5.2 Local freezing and patchwise commuting approximation

Let nonoverlapping patches $\{\Omega_r\}$ partition the spatial grid. On Ω_r , freeze the coefficient at a symmetric positive semidefinite representative block H_r . Let G_r be a symmetric patch regularizer with a specified closure, such as periodic or cosine, and define

$$\tilde{H}_r = I_{|\Omega_r|} \otimes H_r, \quad \tilde{G}_r = G_r \otimes I_d.$$

Then $\tilde{H}_r \tilde{G}_r = \tilde{G}_r \tilde{H}_r$. If $H_r z_j = h_{r,j} z_j$ and $G_r \phi_\xi = g_r(\xi) \phi_\xi$, the legitimate joint modes are $\phi_\xi \otimes z_j$ and their exact local factors are $e(h_{r,j}, g_r(\xi); \rho, \alpha)$.

5.3 Global minimax over local modes

One global pair is selected across all patches; patches do not receive separate ADMM penalties. With attained patch symbol endpoints g_r^- and g_r^+ , set

$$\mathcal{C} = \bigcup_{r,j} \{(h_{r,j}, g_r^-), (h_{r,j}, g_r^+)\}.$$

The patch-surrogate objective is exactly

$$U_{\text{patch}}(\rho, \alpha) = \max_{(h,g) \in \mathcal{C}} |e(h, g; \rho, \alpha)|. \quad (13)$$

The same finite branch argument produces a minimax pair over all local models and $\mathcal{I} \times (0, 2]$.

5.4 Optional variable-coefficient rate bounds

Rigorous noncommuting bounds may be evaluated after tuning. For example, if $\tilde{H} = \bigoplus_r \tilde{H}_r$ and $\tilde{G} = \bigoplus_r \tilde{G}_r$ are the full-size direct-sum surrogates on the patch partition, and

$$\delta_H = \|H - \tilde{H}\|_2, \quad \delta_G = \|G - \tilde{G}\|_2,$$

then the Cayley resolvent identity gives

$$R_{\text{true}}(\rho, \alpha) \leq U_{\text{patch}}(\rho, \alpha) + \frac{\alpha}{\rho} (\delta_H + \delta_G). \quad (14)$$

Each direct-sum block is simultaneously diagonalizable, so $\tilde{E}_{\rho, \alpha}$ is normal and $\|\tilde{E}_{\rho, \alpha}\|_2 = U_{\text{patch}}(\rho, \alpha)$ when the local symbol endpoints used in (13) are attained. Cross-patch regularizer couplings removed by the direct sum are included in δ_G . This bound requires no commutativity of the full operators but is conservative. It and the sharper block-Gershgorin, signed-rectangle, Perron-comparison, and truncated-kernel constructions are detailed in the supplementary information. None is used during deployed parameter selection.

6 Application to Diffeomorphic Image Registration

6.1 Linearized registration model

Let n grid points lie in dimension $d \in \{2, 3\}$, let Δx be the grid spacing, and let $q_i = \nabla I(x_i)$. At the current linearization, $v_i \in \mathbb{R}^d$ is the velocity increment at x_i and $I_t(x_i)$ is the scalar brightness residual. A damped linearized optical-flow subproblem can be written

$$\min_v \frac{\mu}{2} \sum_{i=1}^n (q_i^\top v_i + I_t(x_i))^2 + \frac{\nu}{2} \|v\|_2^2 + \frac{1}{2} v^\top G v, \quad (15)$$

where $\mu > 0$ weights data fidelity, $\nu \geq 0$ is an optional damping weight, and $G \succeq 0$ is the regularization Hessian. Setting $\nu = 0$ gives the undamped model. Splitting a copy $w = v$ places (15) in (1).

6.2 Rank-one data Hessian

The data Hessian is

$$H = \text{diag}(H_1, \dots, H_n), \quad H_i = \nu I_d + \mu q_i q_i^\top. \quad (16)$$

If $q_i \neq 0$, then H_i has eigenvalue ν with multiplicity $d - 1$ and eigenvalue $\nu + \mu \|q_i\|^2$ along q_i . If $q_i = 0$, all d eigenvalues equal ν . Therefore the attained global curvature endpoints are

$$h_- = \nu, \quad h_+ = \nu + \mu \max_i \|q_i\|^2. \quad (17)$$

6.3 Sobolev regularizer and symbol

The componentwise regularizer has

$$G = G_0 \otimes I_d.$$

For a first-order periodic Sobolev discretization with stiffness $\beta > 0$ and zeroth-order weight $\gamma \geq 0$,

$$g(\xi) = \gamma + \frac{4\beta}{(\Delta x)^2} \sum_{\ell=1}^d \sin^2 \left(\frac{\xi_\ell}{2} \right). \quad (18)$$

Thus $g_- = \gamma$. When every Nyquist-frequency corner is present, as on the even periodic grids used here, $g_+ = \gamma + 4\beta d/(\Delta x)^2$; otherwise the exact maximum over the discrete frequency set is used. Higher-order powers use the corresponding powered endpoints.

6.4 Registration-specific four-corner predictor

For a registration patch, choose a representative gradient $\bar{q}_r$ and set

$$H_r = \nu I_d + \mu \bar{q}_r \bar{q}_r^\top.$$

Local freezing at H_r recovers the separable tensor-product structure. The deployed four-corner surrogate uses the family of pixelwise coefficient freezes $H_r = H_i$; equivalently, the representative set includes every q_i . Hence h_+ in (17) is attained, and the rank-one endpoint collapse leaves the four branches

$$(h_-, g_-), \quad (h_-, g_+), \quad (h_+, g_-), \quad (h_+, g_+).$$

If only mean or median patch representatives are retained, these four branches remain a rectangular envelope but need not equal the patch spectrum. For these attained pixelwise endpoints, define $R_{4C} := U_{4C}$ and

$$(\rho_{4C}, \alpha_{4C}) \in \arg \min_{\rho \in \mathcal{I}, 0 < \alpha \leq 2} R_{4C}(\rho, \alpha).$$

Corollary 6.1 (Rank-one frozen-registration reduction). *Assume $d \geq 2$, the coefficient-frozen family contains every H_i , and the common regularizer spectrum attains g_- and g_+ . Also assume $\nu + \gamma > 0$, or remove the common null mode on an explicitly identified quotient subspace. Then the worst contraction factor over that family equals R_{4C} , and Theorem 4.3 returns a global minimizer of this frozen separable surrogate over $\mathcal{I} \times (0, 2]$. For the assembled variable-coefficient operator, the same pair is a predictor rather than a generally optimal pair.*

The four-corner reduction is exact for the separable commuting surrogate; the resulting parameter pair is therefore globally optimal over $\mathcal{I} \times (0, 2]$ for that surrogate, but serves as a predictor of the optimal parameters for the full variable-coefficient registration iteration.

In general, the selected pair need not belong to

$$\arg \min_{\rho \in \mathcal{I}, 0 < \alpha \leq 2} R_{\text{true}}(\rho, \alpha).$$

The predictor ignores global orientation interactions between spatially varying rank-one blocks; the experiments assess its practical transfer.

6.5 One-shot deployment policy

Algorithm 1: One-Shot Four-Corner ADMM Tuning *Inputs:* initial image gradients $\{q_i\}$, weights μ, ν , regularizer parameters and grid, and admissible penalty interval $\mathcal{I}$.

1. From the initial image gradient, compute $h_- = \nu$ and $h_+ = \nu + \mu \max_i \|q_i\|^2$.
2. Compute g_- and g_+ analytically from the regularizer and grid.
3. Form the four endpoint pairs and their rational functions $\theta_c(\rho)$.
4. On the admissible interval, generate the finite algebraic penalty candidate set of Theorem 4.3.
5. At each candidate compute $a(\rho)$, $b(\rho)$, and $\alpha^*(\rho)$ (or an admissible default in the degenerate case); select the pair minimizing R_{4C} .
6. Run inner oADMM with this fixed pair.
7. In one-shot deployment, reuse the pair at later outer registration iterations and pyramid levels unless an explicit retuning policy is enabled.

An *inner ADMM iteration* solves one quadratic problem with fixed H, G, b , so its optimal fixed parameters do not change during that solve. An *outer registration iteration* updates the deformation and relinearizes; the warped image gradients and H may then change. The one-shot policy accepts this model drift to avoid repeated tuning. The reported ablation tests per-pyramid-level retuning, not separate retuning at every outer Gauss–Newton iteration.

7 Experimental Protocol

The reference two-dimensional solver uses a three-level Gaussian pyramid, spatial image gradients, Gauss–Newton optical-flow linearization, Sherman–Morrison solves for the rank-one data blocks, and FFT solves for the Sobolev regularizer. Velocity increments are composed with the displacement. A backtracking line search requires objective decrease and a discrete Jacobian above 0.05. Pyramid transfer is audited and, if needed, its nontranslational component is halved until the same floor holds. All methods in a comparison share initialization, objective, stopping tolerances, line search, pyramid, and topology safeguards.

The spectral tests use direct eigendecomposition only on tractable systems. The controlled spectral studies use $\mathcal{I} = [10^{-2}, 10^2]$; the real-image selector and the 8×8 spectral diagnostic use

$\mathcal{I} = [10^{-4}, 10^3]$. Implementations use $\alpha \in [10^{-6}, 2]$ as a numerical realization of $0 < \alpha \leq 2$. The constant-coefficient test uses periodic $n \times n$ grids, $(h_-, h_+) = (0.2, 2.3)$, and first-order regularization with $(\beta, \gamma) = (0.2, 0.1)$. Controlled noncommuting tests separate periodic magnitude variation, smooth orientation variation, a sharp orientation interface, and Gaussian-smoothed image gradients on 3×3 grids. For these tests, let R_{act} be the true spectral radius at the selected parameters, R_{ora} the radius at numerically oracle-optimized parameters, and U a rigorous enclosure evaluated at the selected pair. We report the parameter-selection gap

$$\frac{R_{\text{act}} - R_{\text{ora}}}{R_{\text{ora}}}$$

and the enclosure gap $(U - R_{\text{act}})/R_{\text{act}}$. Thus the former measures the spectral performance lost by parameter selection, not the Euclidean distance between two parameter vectors.

The real-image pilot uses a public-domain retinal image and an unrestricted immunohistochemistry image under held-out smooth known deformations. CIMA uses the five differently stained sections of the public lung-lesion-3 sample and all ten unordered pairs [3], resized to 256^2 after a common structural-channel preprocessing. FIRE uses all 134 landmarked pairs (14 A, 49 P, and 71 S) at 256^2 [8]. Landmarks are used only for evaluation. Timings use one CPU BLAS/OpenMP thread. Reproducibility details, baseline safeguards, seeds, and certificate protocols are in the supplement.

8 Results

8.1 Exact four-corner validation

This first experiment is a controlled synthetic verification of Theorem 4.2, not an image-registration benchmark. We form a constant-coefficient quadratic split on small periodic $n \times n$ grids with prescribed data-curvature endpoints $(h_-, h_+) = (0.2, 2.3)$ —corresponding to $(\nu, \mu \max_i \|q_i\|^2) = (0.2, 2.1)$ —and a first-order Sobolev regularizer with $(\beta, \gamma) = (0.2, 0.1)$. Because the data coefficient is spatially constant, the operators commute and the joint modal spectrum is exactly the Cartesian product of the two curvature intervals with the discrete regularizer symbol; no image intensity or landmark data enter this test.

The finite algebraic selector returned $(\rho, \alpha) = (0.543512, 1.898152)$ on every grid. Table 1 compares the four-corner envelope U_{4C} with the spectral radius of the full reduced operator $E_{\rho, \alpha}$, obtained by direct eigendecomposition. The two agree to machine precision, confirming exact four-corner equality for this separable model. Panel (a) of Fig. 1 displays the same agreement.

Table 1: Synthetic constant-coefficient verification of exact four-corner equality on periodic grids (no image data).

n	Direct radius $\varrho(E)$	$ U_{4C} - \varrho(E) $
4	0.35313044	$< 10^{-15}$
6	0.35313044	$< 10^{-15}$
8	0.35313044	$< 10^{-15}$

8.2 Controlled noncommuting tests

This experiment remains synthetic: it asks how well the four-corner predictor transfers when the data Hessian varies spatially, so that $HG \neq GH$ and the exact four-corner equality no longer holds.

On 3×3 periodic grids we constructed ten variable-coefficient optical-flow Hessians from magnitude variation, smooth orientation variation, one sharp orientation interface, and Gaussian-smoothed random gradients; nine cases are smooth. For each case we compared the predictor pair with a numerical spectral oracle for the true reduced operator and evaluated optional rate certificates at the selected parameters. No registration images or landmarks are used here.

Across the ten cases, the predictor had a median relative parameter gap of 0.057 overall and 0.027 on the smooth cases, whereas the single abrupt-interface case had a gap of 0.193; no case collapsed to the parameter boundary. At the selected parameters, the median absolute relative discrepancy between R_{4C} and R_{true} was 1.4×10^{-15} and the maximum was 2.8×10^{-3} . The rigorous Perron comparison bound was below one in all ten cases, but its median enclosure gap was 1.29. In a separate 18-case comparison of signed enclosures, every enclosure contained the exact radius. The Perron rectangle was the tightest of the signed constructions, with median enclosure and parameter gaps 0.595 and 0.386, but remained too conservative for online tuning. The detailed certificate table and scaling study are in the supplement.

The archived controlled runs distinguish magnitude, smooth orientation, abrupt-interface, and smoothed-random regimes, but did not store a normalized commutator magnitude. Adding $\|[H, G]\|_2 / (\|H\|_2 \|G\|_2)$ as a horizontal-axis diagnostic is therefore a TODO rather than an unreported result.

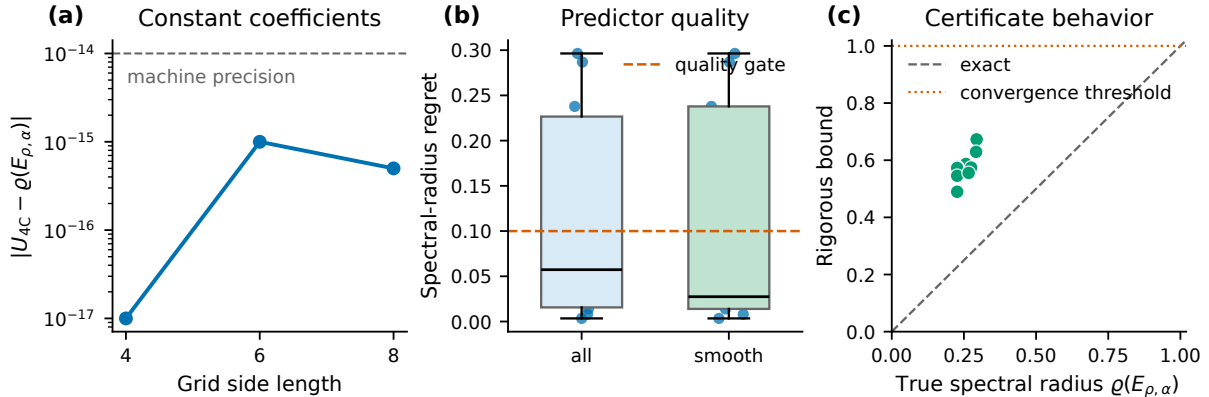


Figure 1: Spectral validation. (a) Exact equality for the separable constant-coefficient model. (b) Spectral-radius regret under controlled variable coefficients; “all” includes the abrupt-interface case. (c) A rigorous but conservative optional certificate.

Controlled real-content pilot. Before the public landmark benchmarks, we ran a small end-to-end registration pilot on real image content with known synthetic warps. A public-domain retinal image and an unrestricted immunohistochemistry image were each subjected to five smooth known deformations; cases 1–2 selected the external fixed pair $(\rho, \alpha) = (0.1, 1)$, and held-out cases 3–5 provided six image–deformation tests. The task is ordinary linearized diffeomorphic registration under the shared solver protocol, measured by inner iterations, runtime, intensity MSE, and discrete Jacobians rather than landmarks. The one-shot predictor reduced median inner iterations by 44.4% and median total time by 41.7% (bootstrap 95% interval $[40.9, 42.6]\%$) relative to that validation-selected fixed pair. Selection consumed 1.58% of proposed-method runtime, median relative MSE change was -0.0068% , and none of the six tested cases had a nonpositive Jacobian. This two-image pilot is a solver check rather than a landmark benchmark.

8.3 Direct spectral optimization comparison

This diagnostic uses the CIMA histology pairs below, but replaces four-corner tuning by a Song-style spectral search on a coarse surrogate. For each pair, the exact variable-coefficient reduced operator was formed on an 8×8 downsampled grid, ρ was optimized numerically, and the resulting pair was transferred to the full 256^2 solve. At each trial penalty, if $\{\lambda_j(\rho)\}$ are the eigenvalues of $C_H(\rho)C_G(\rho)$, the exact conditional relaxation was computed as

$$\alpha_\rho \in \arg \min_{10^{-6} \leq \alpha \leq 2} \max_j \left| 1 - \frac{\alpha}{2} + \frac{\alpha}{2} \lambda_j(\rho) \right|.$$

The question is therefore one of tuning overhead and coarse-model transfer, not of superiority to full-resolution Song optimization. The spectral search required a median 97 spectral-objective evaluations and 2.18 seconds of tuning. Median full-solve iterations were 251, compared with 210 for four-corner tuning, and total times were 4.39 and 2.02 seconds, respectively, with indistinguishable TRE.

Future-work experiment (TODO). On a tractable resolution, directly optimize the true spectral radius over both parameters and report the regret

$$\frac{R_{\text{true}}(\rho_{4C}, \alpha_{4C}) - R_{\text{true}}(\rho^*, \alpha^*)}{R_{\text{true}}(\rho^*, \alpha^*)}.$$

No result for this requested direct-regret experiment is currently reported.

8.4 CIMA histology benchmark

We next turn to real multimodal histology registration. The public lung-lesion-3 sample from the CIMA landmark repository [3] comprises five differently stained sections and all ten unordered pairs. Images supplied at 5% scale were converted to a common structural channel and resized to 256^2 ; landmarks supplied at 50% scale were rescaled for evaluation only. The problem is diffeomorphic registration of these pairs under the shared pyramid solver, comparing one-shot four-corner oADMM tuning with fixed-parameter, residual-balancing, and BB-proxy baselines on target registration error (TRE), iterations, runtime, and discrete Jacobians.

The fixed comparator $(\rho, \alpha) = (0.1, 1)$ was selected on the separate real-content pilot and frozen before CIMA. Across five order-alternated repetitions of all ten pairs (50 paired runs), four-corner tuning reduced median total runtime by 20.1% (bootstrap 95% interval 19.4–20.7%), solver time by 23.5%, and iterations by 30.1%. Selection used 1.16% of proposed runtime. Median TRE was 7.33 pixels (2.02% of the image diagonal) and differed by less than 3×10^{-5} pixels from the fixed comparator. The median landmark-error improvement relative to the unregistered pairs was 36.5%. No global or interior nonpositive Jacobian was observed. One difficult stain pair retained TRE above 22 pixels, identifying initialization and cross-stain representation rather than inner ADMM convergence as the main failure mode.

Table 2: CIMA histology registration at 256^2 : medians of iterations and TRE over ten pairs, and of total time over 50 paired runs.

Method	Iterations	Time (s)	TRE (px)
Fixed ADMM (1, 1)	1910.5	12.09	7.3310
Fixed oADMM (1, 1.8)	1129.5	7.37	7.3308
Residual balancing	1124.5	7.40	7.3310
BB proxy	435.5	3.44	7.3306
External fixed (0.1, 1)	303.5	2.53	7.3306
Four-corner one-shot	210.0	2.02	7.3306

8.5 FIRE retinal benchmark

The main end-to-end study uses the public FIRE fundus registration benchmark [8]: all 134 landmarked pairs (14 A, 49 P, and 71 S), converted to a foreground-percentile-normalized green channel and resized to 256^2 . Landmarks are used only for evaluation. The problem is retinal diffeomorphic registration under identical phase-correlation initialization, pyramid, objective, and stopping criteria, with one CPU BLAS/OpenMP thread. One-shot four-corner tuning is compared with the external fixed pair, residual balancing, a safeguarded BB proxy, and fixed oADMM on iterations, runtime, TRE, and topology.

Across all pairs, median iterations were 245 for one-shot four-corner tuning and 412 for the external fixed pair, a paired median reduction of 39.6%. Median total times were 4.774 and 6.680 seconds, a 34.4% paired reduction (bootstrap 95% interval 30.7–35.7%); tuning accounted for 1.2% of proposed total time. The safeguarded BB proxy required 623.5 iterations and 9.435 seconds; the predictor was faster on 94.0% of pairs, with a median paired reduction of 51.7%. Median TRE differed from the fixed pair by -1.4×10^{-5} pixels, the maximum absolute difference was 3.9×10^{-4} pixels, and no method produced a nonpositive Jacobian.

Table 3: FIRE retinal registration: 134 pairs at 256^2 under identical initialization and solver protocol.

Method	Iter.	Time (s)	TRE (px)	TRE/diag. (%)
External fixed (0.1, 1.0)	412.0	6.680	5.50178	1.51967
Residual balancing	1434.0	22.325	5.50211	1.51976
BB proxy	623.5	9.435	5.50175	1.51966
Fixed oADMM (1.0, 1.8)	1529.5	23.471	5.50194	1.51971
Four-corner one-shot	245.0	4.774	5.50177	1.51966

Pairwise percentage reductions need not equal percentage differences between separately reported method medians.

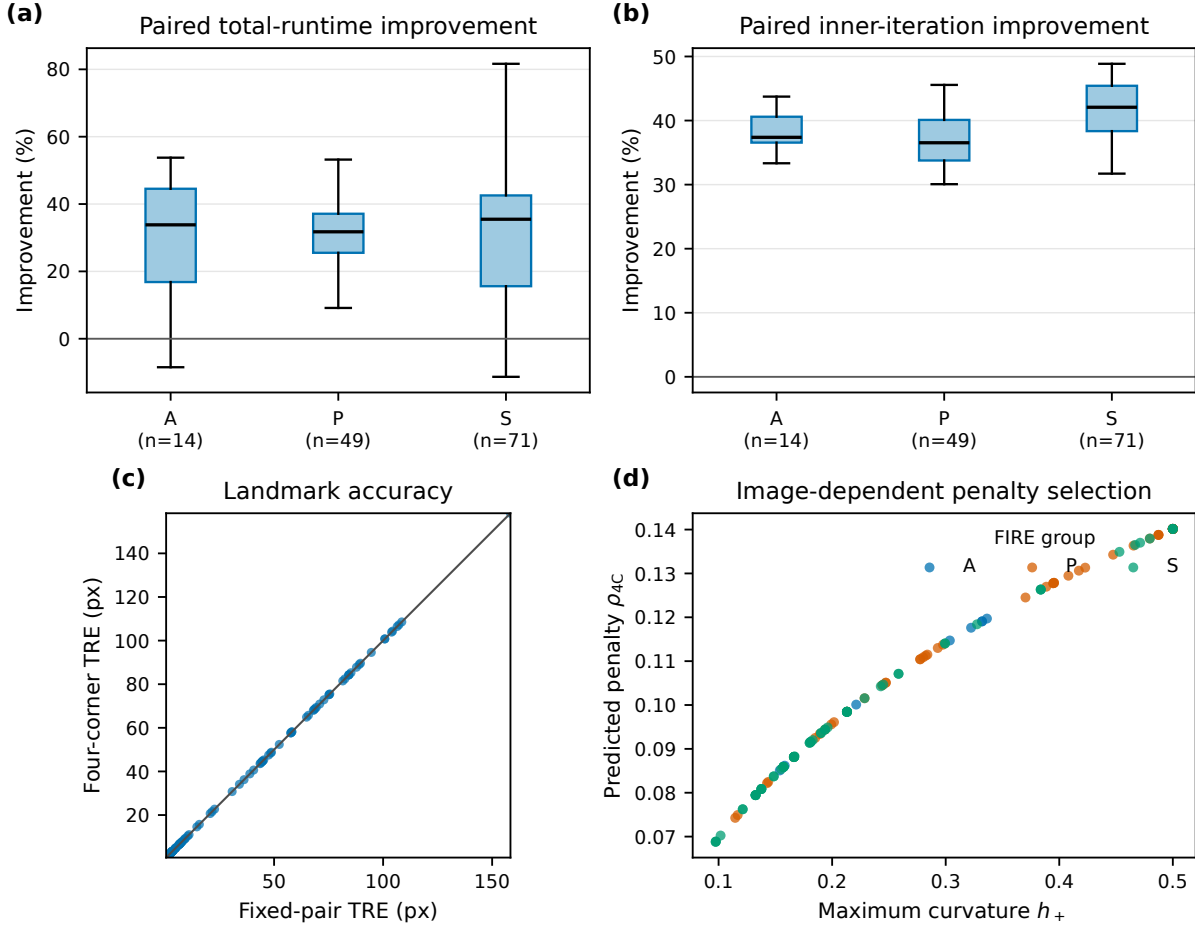


Figure 2: FIRE at 256×256 . (a) Pairwise total-runtime reduction relative to the external fixed pair. (b) Inner-iteration reduction. (c) TRE agreement. (d) Image-dependent predicted penalty.

8.6 One-shot reuse ablation

This ablation stays within FIRE but isolates the parameter-reuse policy. On a balanced 18-pair FIRE subset we compared the deployed one-shot policy—one pair from the initial full-resolution gradient, reused at every pyramid level—with retuning at every pyramid level. The question is whether fewer inner iterations from retuning reduce end-to-end runtime once tuning cost is included. Retuning reduced median inner iterations from 243 to 222 but increased median total time from 2.651 to 3.895 seconds; median tuning time also increased from 0.030 to 0.110 seconds. The archived aggregate timings do not isolate the remainder of the increase, but they establish that fewer iterations did not imply lower end-to-end runtime in this per-level pipeline. The reported deployment policy therefore reuses one pair; per-outer-iteration retuning was not separately tested.

9 Discussion

9.1 Why four corners work

The reduction follows from two facts: the separable model supplies independent data and regularizer modal coordinates, and the sign of each partial derivative in (9) is independent of the differentiated

coordinate. Convexity after the affine relaxation map then preserves endpoint attainment. This argument is independent of image size.

9.2 Exact model optimum versus full-problem prediction

Exact surrogate optimization is not exact variable-coefficient optimization. The local frozen model has a Cartesian-product spectrum, but full registration generally has $HG \neq GH$. The practical pair is therefore surrogate-optimal and predictive for the full iteration. Its strong empirical performance suggests that endpoint curvature information captures useful scale balance, especially for smooth coefficients, but it is not a theorem about the minimizer of R_{true} .

9.3 Comparison with Song et al.

Song et al. address a more general problem class [12]. Their parameter optimization pays for repeated spectral-radius evaluation while finding ρ ; neither their method nor ours requires changing ρ during a fixed inner solve. Four-corner tuning avoids this search under stronger separable assumptions and becomes a predictor outside them. The 8×8 diagnostic quantifies overhead and transfer only.

9.4 Limitations

Local freezing is approximate, and the rank-one endpoint collapse omits orientation interactions. The optional noncommuting certificates discard directional cancellation and are conservative. CIMA contains one five-image tissue set at reduced resolution; FIRE is evaluated at 256^2 under a CPU protocol and against an external fixed comparator, residual balancing, and a safeguarded BB proxy. The BB proxy does not reproduce every adaptive ADMM variant. The direct true-spectral-radius regret experiment remains TODO. Finally, the theory describes each convex linearized subproblem, not global convergence of the outer nonlinear diffeomorphic registration scheme.

10 Conclusion

For quadratic oADMM with an attained Cartesian-product joint modal spectrum, the worst contraction is exactly one of four endpoint factors. Conditional relaxation and a finite algebraic penalty set then give search-free joint tuning. Spatially varying registration is noncommuting, but local freezing and its rank-one Hessian yield an inexpensive one-shot predictor. The FIRE and CIMA results show substantial iteration and runtime reductions at matched accuracy and observed topology. Future work should quantify direct spectral-radius regret at tractable resolutions, extend the method to volumetric registration, and tighten optional noncommuting bounds.

Declarations

- **Funding.** No specific funding was received for this work.
- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** Not applicable; the study uses only publicly available, de-identified image datasets.
- **Consent for publication.** Not applicable.

- **Data availability.** FIRE [8] and CIMA [3] are publicly available from their original providers. The FIRE archive is not redistributed.
- **Materials availability.** Not applicable.
- **Code availability.** Source code, tests, configurations, processed JSON and CSV outputs, and preparation and evaluation scripts are available at <https://github.com/xhxuciedu/admm-registration>.
- **Author contributions.** Katherine Xie and Jie Wu conducted the research and wrote and edited the manuscript. Xiaohui Xie conceived the work and edited the manuscript.

A Proof of the Reduced Cayley Representation

Set $P_H = \rho(H + \rho I)^{-1}$ and $P_G = \rho(G + \rho I)^{-1}$. For standard ADMM, the homogeneous state map is

$$\begin{bmatrix} w^{k+1} \\ u^{k+1} \end{bmatrix} = \underbrace{\begin{bmatrix} P_G \\ I - P_G \end{bmatrix}}_A \underbrace{\begin{bmatrix} P_H & I - P_H \end{bmatrix}}_B \begin{bmatrix} w^k \\ u^k \end{bmatrix}.$$

The nonzero eigenvalues of AB equal those of

$$BA = P_H P_G + (I - P_H)(I - P_G).$$

Since $P_H = (I - C_H)/2$ and $P_G = (I - C_G)/2$,

$$BA = \frac{1}{2}(I + C_H C_G).$$

Inserting (5) makes the reduced relaxed map $(1 - \alpha)I + \alpha BA$, which is (6).

B Proof Details for the Four-Corner Theorem

For fixed g ,

$$\theta(h, g; \rho) = \frac{\rho^2 + hg}{(\rho + g)(\rho + h)}$$

has derivative given by (9), whose sign is that of $g - \rho$ and is constant in h . The analogous statement holds after interchanging h and g . Repeated one-dimensional endpoint maximization therefore puts both the minimum and maximum of θ at rectangle corners. Because every value of θ lies between those extrema and absolute value of an affine function is convex, no interior value can exceed both endpoint images. Equality with the operator radius additionally requires that the maximizing endpoint pair be an actual common mode. Definition 4.1 and attained endpoints guarantee all four such modes.

C Proof of Finite Algebraic Selection

Write

$$\theta_c(\rho) = \frac{\rho^2 + h_c g_c}{(\rho + h_c)(\rho + g_c)}.$$

Identical rational branches are first deduplicated. Every denominator is positive for $\rho > 0$. Pairwise equality of two distinct branches, a transition equality $\theta_i + \theta_j = 1$, and stationarity of either branch pair in (12) therefore become polynomial equations after denominators are cleared. Direct expansion gives degrees at most three, four, and six, respectively. Identically zero equations are discarded; if an active reduced objective is constant on a piece, its closure endpoints represent all of its minimizers needed by the finite candidate set.

Sort the interval endpoints and all branch intersections. On each resulting open interval, one branch realizes a and one realizes b . The reduced objective is a smooth rational function except where $a + b = 1$. A minimum on the closure of each piece is consequently attained at an endpoint, at that transition, or at a stationary point. Taking the union over all branch pairs and retaining roots that satisfy interval membership and branch activity gives the finite superset in Theorem 4.3. Evaluating the original four-corner objective filters extraneous roots introduced by denominator clearing.

References

- [1] J. Ashburner, “A fast diffeomorphic image registration algorithm,” *NeuroImage*, 38(1):95–113, 2007.
- [2] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, “Distributed optimization and statistical learning via the alternating direction method of multipliers,” *Foundations and Trends in Machine Learning*, 3(1):1–122, 2011.
- [3] J. Borovec, A. Muñoz-Barrutia, and J. Kybic, “Benchmarking of image registration methods for differently stained histological slides,” in *IEEE ICIP*, pp. 3368–3372, 2018.
- [4] T. F. Chan and H. C. Elman, “Fourier analysis of iterative methods for elliptic problems,” *SIAM Review*, 31(1):20–49, 1989.
- [5] E. Ghadimi, A. Teixeira, I. Shames, and M. Johansson, “Optimal parameter selection for ADMM: quadratic problems,” *IEEE Transactions on Automatic Control*, 60(3):644–658, 2015.
- [6] P. Giselsson and S. Boyd, “Linear convergence and metric selection for Douglas–Rachford splitting and ADMM,” *IEEE Transactions on Automatic Control*, 62(2):532–544, 2017.
- [7] A. Hering, L. Hansen, T. C. W. Mok, et al., “Learn2Reg: comprehensive multi-task medical image registration challenge,” *IEEE Transactions on Medical Imaging*, 42(3):697–712, 2023.
- [8] C. Hernández-Matas, X. Zabulis, A. Triantafyllou, P. Anyfanti, S. Douma, and A. A. Argyros, “FIRE: fundus image registration dataset,” *Journal for Modeling in Ophthalmology*, 1(4):16–28, 2017.
- [9] P. Kumar, C. Rodrigo, F. J. Gaspar, and C. W. Oosterlee, “On local Fourier analysis of multigrid methods for PDEs with jumping and random coefficients,” *SIAM Journal on Scientific Computing*, 41(3):A1385–A1413, 2019.
- [10] A. Mang and G. Biros, “An inexact Newton–Krylov algorithm for constrained diffeomorphic image registration,” *SIAM Journal on Imaging Sciences*, 8(2):1030–1069, 2015.
- [11] C. Rodrigo, F. J. Gaspar, and L. T. Zikatanov, “On the validity of the local Fourier analysis,” arXiv:1710.00408, 2017.

- [12] J. Song, W. Lu, Y. Lei, Y. Tang, Z. Pan, and J. Duan, “Optimizing ADMM and over-relaxed ADMM parameters for linear quadratic problems,” in *Proceedings of AAAI*, 38(8):8117–8125, 2024.
- [13] A. Teixeira, E. Ghadimi, I. Shames, H. Sandberg, and M. Johansson, “The ADMM algorithm for distributed quadratic problems: parameter selection and constraint preconditioning,” *IEEE Transactions on Signal Processing*, 64(2):290–305, 2016.
- [14] A. Thorley, X. Jia, H. J. Chang, et al., “Nesterov accelerated ADMM for fast diffeomorphic image registration,” in *MICCAI*, LNCS 12904, pp. 150–160, 2021.
- [15] T. Vercauteren, X. Pennec, A. Perchant, and N. Ayache, “Diffeomorphic demons: efficient non-parametric image registration,” *NeuroImage*, 45(1):S61–S72, 2009.
- [16] R. Wienands and W. Joppich, *Practical Fourier Analysis for Multigrid Methods*. Chapman & Hall/CRC, 2005.
- [17] B. Wohlberg, “ADMM penalty parameter selection by residual balancing,” arXiv:1704.06209, 2017.
- [18] Z. Xu, M. A. T. Figueiredo, and T. Goldstein, “Adaptive ADMM with spectral penalty parameter selection,” in *AISTATS*, PMLR 54:718–727, 2017.